Curriculum Vitae

Edmilson Roque dos Santos

Research interests: Dynamical Systems, Ergodic Theory, Synchronization, Sparse Recovery methods, Reservoir Computing.

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https://github.com/edmilson-roque-santos

Professional Appointments

Sep 2025 – to date

Postdoctoral Researcher
Nonlinear Dynamics and Time Series analysis
Max Planck Institute for the Physics of Complex Systems
Project: *Learning sparse network dynamics from data*

Feb 2023 – May, 2025

Research Associate
Clarkson Center for Complex Systems Science
Clarkson University

NSF-NIH CRCNS: *Functional Brain Networks with Tensioned Stability for Optimal Processing*

Education

Feb 2018 – Jan 2024

PhD. in Applied Mathematics
ICMC-USP and Imperial College London (partially supported by Royal Society).

Project Title: *Reconstruction of Complex Networks from Data*
Supervisor: Prof. Dr. Tiago Pereira
Co-Supervisor: Prof. Dr. Sebastian van Strien

Feb. 2016 – Feb. 2018

MSc. in Physics
IFSC-USP

Thesis title: *Discontinuous transitions to collective dynamics in star motifs of coupled oscillators*
Supervisor: Prof. Dr. Tiago Pereira
Co-Supervisor: Dr. Jaap Eldering

Feb. 2012 – Dec. 2015

Bachelor in Physics
IFSC-USP

Undergraduate research title: *Models in Explosive Synchronization*
Supervisor: Prof. Dr. Francisco Aparecido Rodrigues

Publications

Preprints and in preparation

1. Edmilson Roque dos Santos. *"Learning sparse network dynamics from transient macroscopic measurements"*. In preparation.
2. Daniel Estevez Moya, Francesco Martinuzzi, Edmilson Roque dos Santos, Erick Alejandro Madrigal Solis, Ernesto Estevez Rams, and Holger Kantz. *"Characterizing the better performance of multi-model ensemble forecasts through reservoir computing"*. In preparation.
3. Edmilson Roque dos Santos. *Reconstruction of bursting network dynamics from data.* Under review.
4. Edmilson Roque dos Santos, Sebastian van Strien, and Tiago Pereira. *"Ergodic Basis Pursuit induces Divide-and-Conquer Network Reconstruction"*. In preparation.

Journal Publications

1. Edmilson Roque dos Santos and Erik Bollt. *"On the emergence of numerical instabilities in Next Generation Reservoir Computing"*. https://doi.org/10.1063/5.0278709. Chaos 35, 123102 (2025).

2. Yuanzhao Zhang, Edmilson Roque dos Santos, Huixin Zhang, and Sean P. Cornelius. *"How more data can hurt: Instability and regularization in next-generation reservoir computing"*. https://doi.org/10.1063/5.0262977. Chaos 35, 073102 (2025).

3. Tiago Pereira, Edmilson Roque dos Santos, Sebastian van Strien. *"Robust reconstruction of sparse network dynamics"*. https://iopscience.iop.org/article/10.1088/1361-6544/add3b0, Nonlinearity 38 055031 (2025).

4. Anil Kumar, Edmilson Roque dos Santos, Paul J. Laurienti, and Erik Bollt. *"Symmetry breaker governs synchrony patterns in neuronal inspired networks"*. Chaos 34, 113115 (2024). https://doi.org/10.1063/5.0209865

5. Erik Bollt, Jeremie Fish, Anil Kumar, Edmilson Roque dos Santos, and Paul J. Laurienti. *"Fractal Basins as a Mechanism for the Nimble Brain"*. Sci Rep 13, 20860 (2023). https://doi.org/10.1038/s41598-023-45664-5.

6. Juliano Genari, Guilherme T. Goedert, Sérgio H.A. Lira, Krerley Oliveira, Adriano Barbosa, et al. "Quantifying protocols for safe school activities". PLoS ONE 17(9): e0273425 (2022). https://doi.org/10.1371/journal.pone.0273425

7. Marcel Novaes, Edmilson Roque dos Santos, Tiago Pereira. *"Recovering sparse networks: Basis adaptation and stability under extensions"*. Physica D: Nonlinear Phenomena 424 132895, (2021). https://doi.org/10.1016/j.physd.2021.132895

8. Jaap Eldering, Jeroen Lamb, Tiago Pereira, Edmilson Roque dos Santos. *"Chimera states through invariant manifold theory"*. Nonlinearity 34-5344, (2021). https://dx.doi.org/10.1088/1361-6544/ac0613

Grants and Honors

July, 2025 - to date — EPSRC-FAPESP predicting critical transitions in complex dynamical networks: reduction and learning.
Affiliated with the project led by Tiago Pereira (USP, Brazil) and Jeroen Lamb (Imperial College London, UK)

May, 2025 — Life Sciences and Dynamical Systems Travel Fund by Dr. Simone Bianco
SIAM Travel Awards

2018 - 2022 — Reconstruction of complex networks from data
Doctoral Scholarship: The São Paulo Research Foundation, FAPESP

2016 - 2018 — Grant: Coordination for the Improvement of Higher Education Personnel, CAPES.

2015 — Garfield's Medal: Best Oral Presentation in the Symposium of Mathematics for the Undergraduate course (SiM 2015) at ICMC-USP.

July 2015 — A general analysis of synchronization process
Research internship at PIK under the supervision of Prof. Jurgen Kurths - The São Paulo Research Foundation, FAPESP.

2014 - 2015 — Explosive synchronization models in complex networks
Undergraduate Scientific Initiation Scholarship: The São Paulo Research Foundation, FAPESP.

Conferences and Invited talks

Apr 27, 2026 — PSU-Purdue-UMD Joint Seminar on Mathematical Data Science. *Learning network dynamics from data through compressive sensing techniques.* (Invited talk)

Apr 20 - 24, 2026 — One-Dimensional Dynamics and High-Dimensional Network Systems, Palazzo del Castelletto, Pisa. *Learning network dynamics from data through compressive sensing techniques.* (Invited talk)

Mar 8 - 13, 2026 — DPG meeting - Statistical Physics and Nonlinear Dynamics - Dresden. *Learning spatiotemporal patterns from mean-field data.* (Contributed talk)

Jan 12-13, 2026 — International Workshop: Recent Trends in Coupled Network Systems January 12 - 13, 2026 — Berlin, WIAS. *Learning spatiotemporal patterns from mean-field data.* (Poster presentation)

Oct 9, 2025 — Math Seminar at Center for Systems Biology Dresden. *Ergodic Basis Pursuit leads to exact reconstruction of sparse network dynamics.* (Invited talk)

May 11, 2025 — MS41 - Invariant Sets in Dynamics: Applications and Future Directions. SIAM Conference on Applications of Dynamical Systems 2025. *Switching Between Multiple Invariant Sets in Neuronal-Inspired Network Dynamics.* (Invited talk)

Dec 18, 2024 — (Network) Dynamical Systems. University of São Paulo, São Carlos. *Symmetry breaker governs synchrony patterns in neuronal inspired networks.* (Contributed talk)

Oct 7, 2024 — Applied Math Seminars. University of Ottawa, Canada. *Metastability of chimeras states in coupled networks.* (Invited talk)

July 8 - 12, 2024 — Fourth Symposium of Machine Learning on Dynamical Systems. Fields Institute, Toronto - Canada *Ergodic Basis Pursuit induces exact (and robust) sparse network reconstruction.* (Contributed talk)

Mar 20 - 22, 2024 — NERCCS 2024: Seventh Northeast Regional Conference on Complex Systems. Potsdam, NY - USA. *Dynamics of synchrony patterns on networks.* (Contributed talk)

Feb 5, 2024 — Oberseminar Dynamics. TUM, Munich - Germany. *Ergodic Basis Pursuit induces robust reconstruction of sparse network dynamics.* (invited talk - online format)

Oct 27, 2023 — C3S2 Seminars. The Clarkson Center for Complex Systems Science, Potsdam, NY - USA. *Reconstruction of coupled sparse networks from data.* (invited talk)

Mar 22 - 24, 2023 — NERCCS 2023: Sixth Northeast Regional Conference on Complex Systems. *Ergodic Basis Pursuit induces robust network reconstruction.* (contributed talk)

Jan 9 - 11, 2023 — Dynamics Days US 23. *Ergodic basis pursuit induces robust reconstruction of weakly coupled sparse networks. (contributed talk - online format)*

Dec 17 - 18, 2022 — Mathematical Physics Days 2022. *Reconstruction of Weakly Coupled Sparse Networks from Data. (invited talk - online format)*

Sep 12 - 21, 2022 — Inverse Network Dynamics - NETDAT22. MPI for the Physics of Complex Systems, Dresden - Germany. *Ergodic basis pursuit induces robust reconstruction of sparse networks.* (contributed talk)

Jun 20 - 23, 2022 — Rényi 100. Hungarian Academy of Sciences, Budapest - Hungary. *Ergodic basis pursuit induces robust reconstruction of sparse networks.* (poster presentation)

May 6, 2022 — Free University of Berlin (FUB). Berlin - Germany. *Ergodic basis pursuit induces robust reconstruction of sparse networks.* (presentation)

May 5, 2022 — Potsdam Institute for Climate Impact Research (PIK). Potsdam - Germany. *Ergodic basis pursuit induces robust reconstruction of sparse networks.* (presentation)

May 5, 2022 — Weierstrass Institute for Applied Analysis and Stochastic (WIAS). Berlim - Germany. *Ergodic basis pursuit induces robust reconstruction of sparse networks.* (presentation)

May 4, 2022 — University of Potsdam. Potsdam - Germany. *Ergodic basis pursuit induces robust reconstruction of sparse networks.* (presentation)

May 23 - 27, 2021 — SIAM Conference on Applications of Dynamical Systems (DS21). *Ergodicity implies stable reconstruction of sparse network dynamics.* (invited talk - online format)

April 29, 2021 — Dynamical Systems and Networks Seminars. Courant Institute of Mathematical Sciences, New York - USA. *Chimera states through invariant manifold theory.* (invited talk - online format)

Oct 07 - 11, 2019 — V Escola Brasileira de Sistemas Dinâmicos. UFMG - Belo Horizonte, MG - Brazil.
Chimera states through invariant manifold theory. (poster presentation)

Aug 26 - Sep 01, 2018 — V Workshop and School on Dynamics, Transport and Control in Complex Networks - ComplexNet. INPE, Cachoeira Paulista, SP - Brazil.
Discontinuous transitions to collective dynamics in star motifs of coupled oscillators. (poster presentation)

Jul 27 - 31, 2015 — International Workshop on Dynamics of Coupled Oscillators: 40 years of the Kuramoto Model. MPI for the Physics of Complex Systems, Dresden - Germany.
Title: *Influence of frequency distribution on the discontinuous phase transition in networks of Kuramoto oscillators.* (poster presentation)

Sep 17 - 19, 2014 — Undergraduate Research Project Highlights 22° SIICUSP -The University of São Paulo's International Symposium of Undergraduate Research (SIICUSP) São Paulo - Brazil.
Title: *Influence of frequency distribution on the discontinuous phase transition in networks of Kuramoto oscillators.* (poster presentation)

Oct 6 - 11, 2014 — III Workshop and School on Dynamics, Transport and Control in Complex Networks - ComplexNet. São José dos Campos, SP - Brazil.
Title: *Influence of frequency distribution on the discontinuous phase transition in networks of Kuramoto oscillators.* (contributed talk)

Visiting

Jul - Aug, 2025 — University of São Paulo, ICMC-USP, Brazil
Prof. Dr. Tiago Pereira, Dr. Thomas Peron and Dr. Eddie Nijholt

Mar - Jun 2022 — Imperial College London, London, UK.
Prof. Dr. Sebastian van Strien and Prof. Dr. Jeroen Lamb

Dec 2019 - Mar 2020 — Imperial College London, London, UK.
Prof. Dr. Sebastian van Strien and Prof. Dr. Jeroen Lamb

Jan - May, 2019 — Imperial College London, London, UK.
Prof. Dr. Sebastian van Strien and Prof. Dr. Jeroen Lamb

Jan - Feb, 2018 — Imperial College London, London, UK.
Prof. Dr. Jeroen Lamb

Jul - Aug, 2015 — Potsdam Institute for Climate Impact Research - PIK, Potsdam, Germany
Prof. Dr. Jurgen Kurths

Organizing

Jul 19 - 24, 2026 — Minisymposium at Dynamics Days - Europe 2026 on *Advances in Reconstruction of Network Dynamics from Time Series*
Co-organization with Bengi Dönmez

Jun 28 - Jul 3, 2026 — Half a Century of the Kuramoto Model: Explaining Emergent Order in Time and Space
Co-organization with Dr. Simona Olmi, Prof. Dr. Michael Rosenblum, and Prof. Dr. Marc Timme

Jan, 19-20, 2026 — Mini workshop on Reservoir Computing for Nonlinear Dynamics
Small workshop at MPI-PKS

Oct, 2025 - ongoing — Nonlinear time series analysis seminars at MPI-PKS
Co-organization with Prof. Dr. Holger Kantz

Mar 20 - 22, 2024 — NERCCS 2024
Co-organization with Prof. Dr. Chunlei Liang, Prof. Dr. Erik Bollt, Dr. Golshan Madraki, and Dr. Jeremie Fish

2022 - 2023 — Dynamical systems research seminars at ICMC-USP
Co-organization with Dr. Zheng Bian

Jun - Jul, 2022 — Dynamical systems research seminars
Organization of seminars at Imperial College London

Apr - Dec, 2021 — NetDynamics Seminars between ICMC-USP and Nodds Lab – Dr. Deniz Eroglu's group from Kadir Has University (KHAS)
Co-organization with Dr. Elif Yunt.

Oct 24 - 26, 2018 — São Paulo Dynamical Systems days
Co-organization with Prof. Dr. Tiago Pereira and Prof. Dr. Ali Tahzibi

2014 — SIFSC 4 - Semana Integrada de Física de São Carlos
Co-organization with undergraduate and graduate students from IFSC-USP

2013 — SIFSC 3 - Semana Integrada de Física de São Carlos
Co-organization with undergraduate and graduate students from IFSC-USP

Peer review

Physical Review Letters; Physical Review Applied; Physical Review Research; Chaos: An Interdisciplinary Journal of Nonlinear Science; Physical Review E; Physica A: Statistical Mechanics and its Applications; Physica D: Nonlinear Phenomena; and Chaos, Solitons and Fractals;

Programming languages

Advanced *Python*
Basic *C++, Fortran, Matlab, Julia, Mathematica*

Teaching

2019 — TA for Linear Algebra and Ordinary Differential Equations
University of São Paulo
Undergraduate Course

2017 — TA for Advanced Laboratory of Physics
University of São Paulo
Undergraduate Course

2017 — TA for Mathematical Physics
University of São Paulo
Undergraduate Course

Extra Activities

2020 - 2022 — Modcovid19.
Participation in a large collaboration group formed by different Brazilian institutions to model COVID-19 in Brazil, in particular, model validation of COMORBUSS software, which can be accessed in the following link:

`https://comorbuss.org/Home`.

2014 — Judge during the finals in Brazil.
International Young Physicists Tournament (IYPT)